K223: Nuclear $\gamma\gamma$ Angular Correlations

Dhruv Pisharody[1*] and Angelo Brade[1*]

[1*]Rheinische Friedrich–Wilhelms–Universität, Bonn.

*Corresponding author(s). E-mail(s): ; s72abrad@uni-bonn.de;

1 Analysis of the Data

1.1 *Calibration of the SCAs*

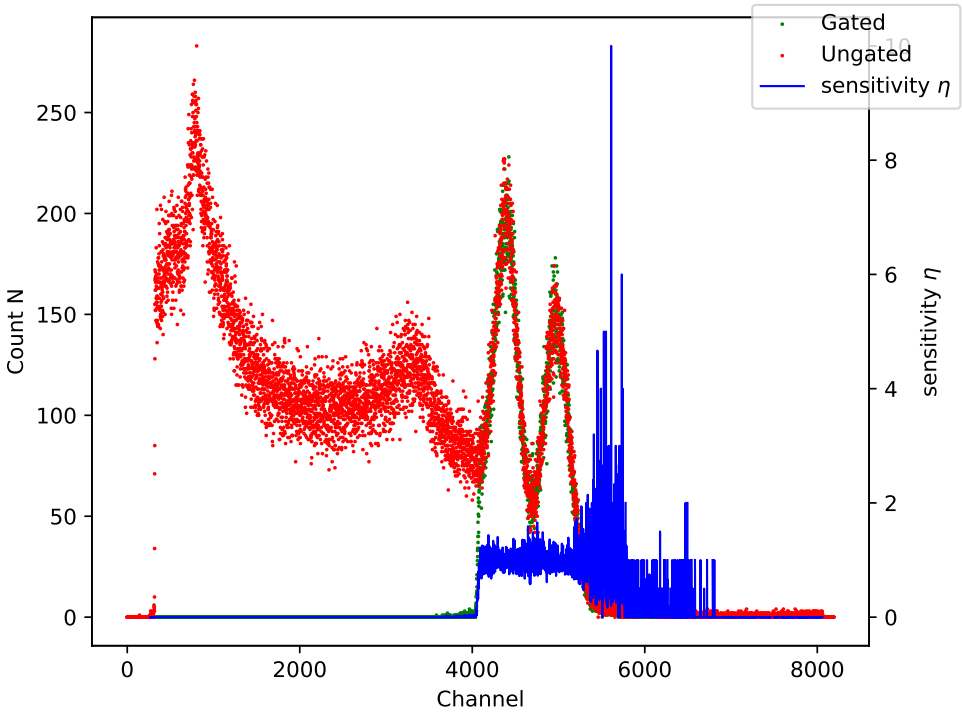

Figure 1: SCA Spectrum for Stationary Detector

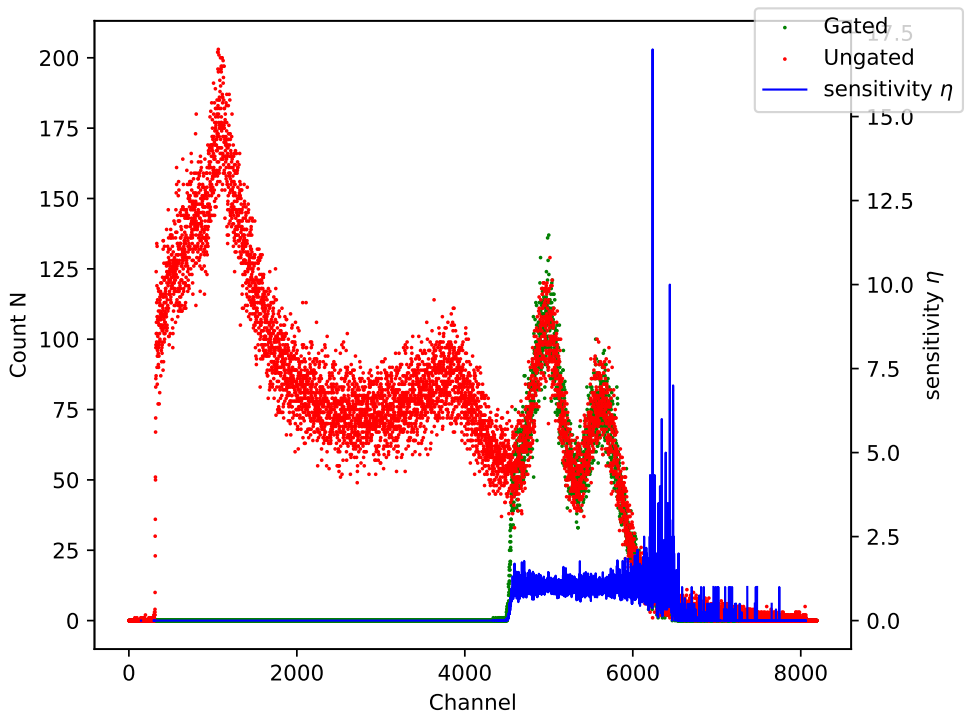

Figure 2: SCA Spectrum for Moving Detector

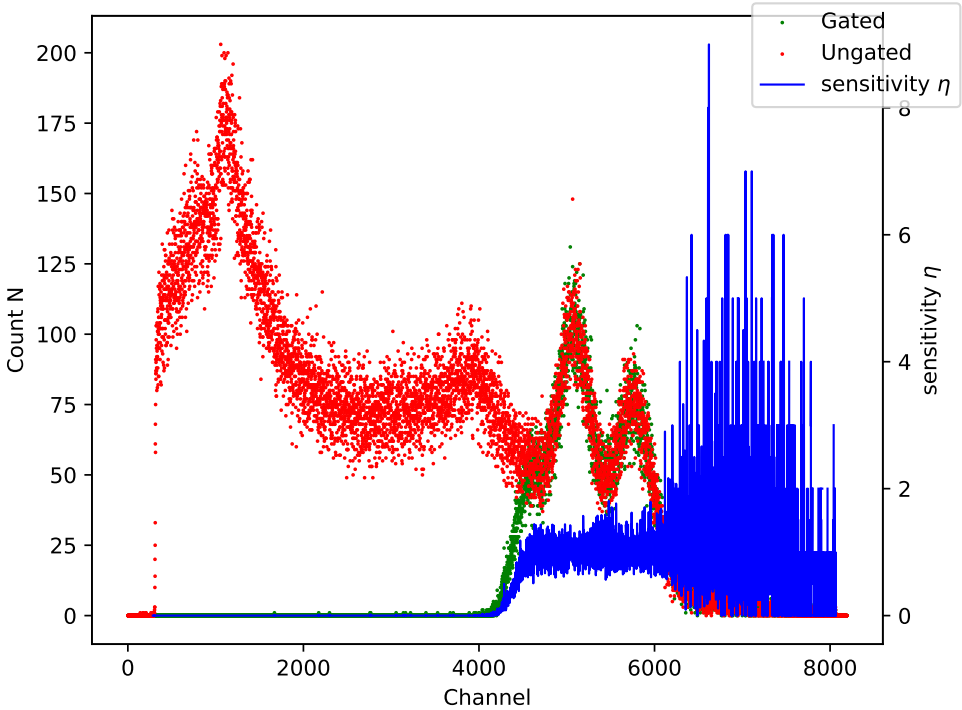

Figure 3: CFD Spectrum for Stationary Detector

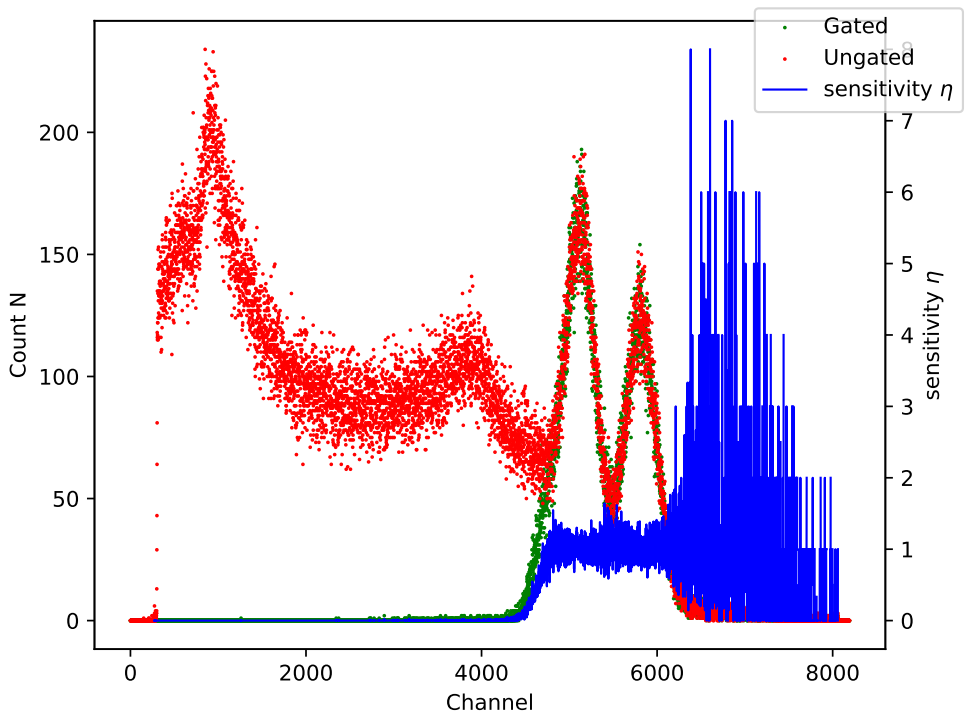

Figure 4: CFD Spectrum for Moving Detector

Figure 5: CFD Spectrum for Moving Detector

Figure 6: CFD Spectrum for Moving Detector

Figure 7: CFD Spectrum for Moving Detector

Figure 8: CFD Spectrum for Moving Detector

Figure 9: CFD Spectrum for Moving Detector

Figure 10: CFD Spectrum for Moving Detector

Figure 11: CFD Spectrum for Moving Detector

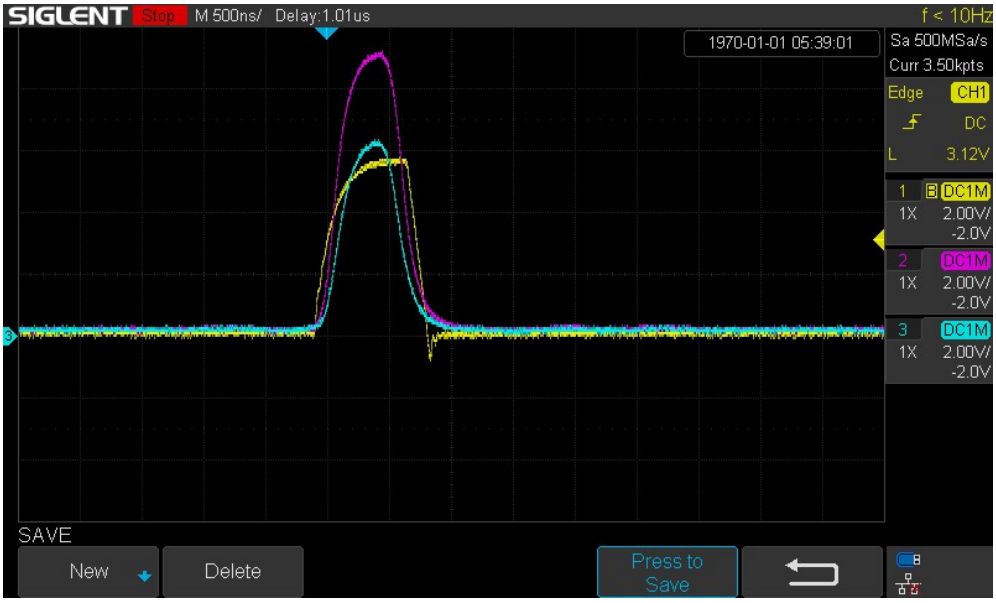

Figure 12: CFD Spectrum for Moving Detector

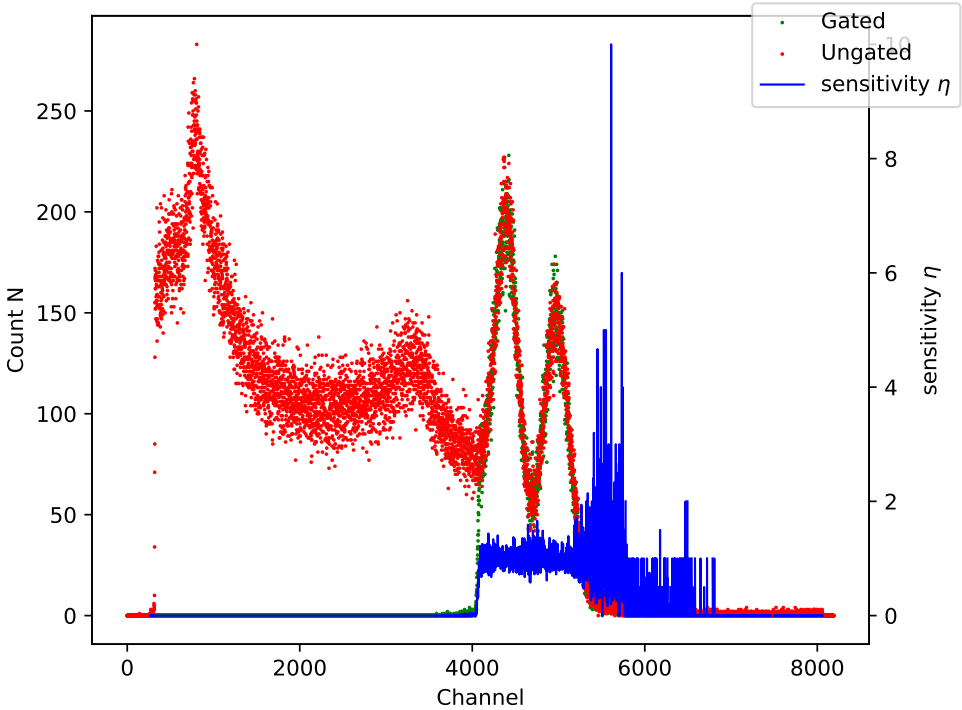

Figure 13: CFD Spectrum for Moving Detector

2 Main Analysis

With the constructed circuit the random measurement and main measurement are taken. For the random measurements a total of 13 angles ranging from 90° to 270° in 15° steps are taken. At each angle the count coincidence rate is measured for 10 Minutes. Since the random measurement is done in 2 runs, every angles was measured twice. For the main measurement the total runs were increased from 2 to 6 and the measuring time per angles per run was increased from 10 Minutes to 12 Minutes.

2.1 *Random Measurements*

2.1.1 *Direct Measurement*

To investigate the random coincidences of the signals, the delay in the fast branch is set to maximal difference to ensure no true coincidence is at play. Now only accidental overlapping signals trigger a coincidence. The rate of random coincidences can be determined by two methods. First, the gated counts are taken. Those already are the random coincidences since only when the signals of the fast branches overlap, a signal from the slow branch is received. As for all measured counts from the spectrum, a

Poisson distribution is assumed, such that the uncertainty is

$$\sigma_N = \sqrt{N} \tag{1}$$

Suppose a count of 0 is registered. That would have yield an uncertainty of also 0. Since an uncertainty of zero is not realistic, a conservative correction is taken such that an uncertainty of

$$\sigma_N = \sqrt{N+1} \tag{2}$$

is assumed. This yields the count rate distribution shown in Figure 14

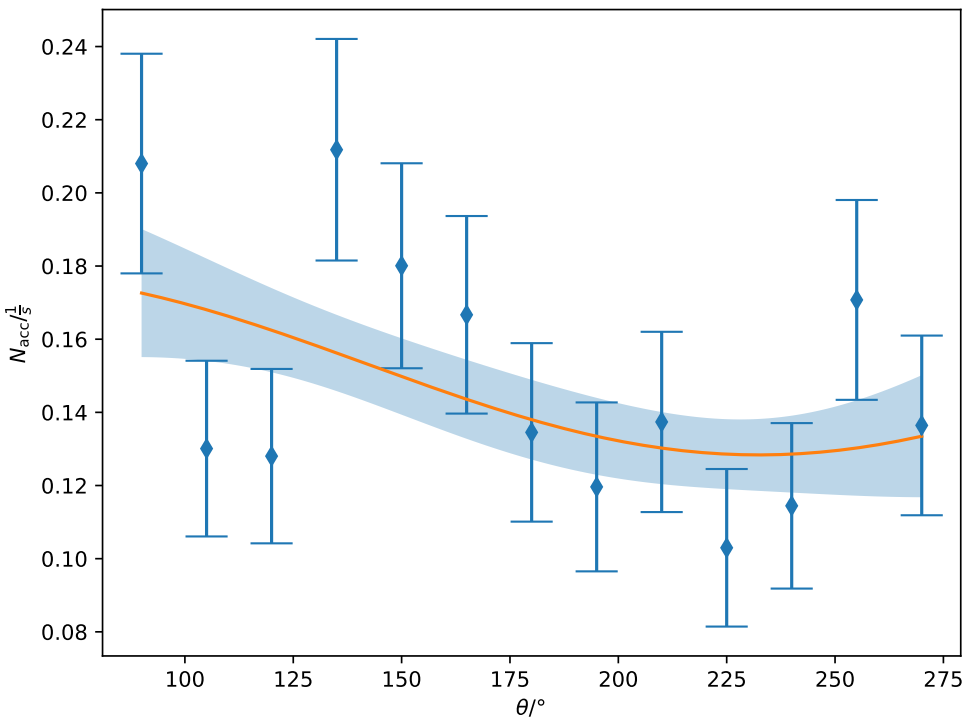

Figure 14: Direct Measurement of Random Coincidences from

It can be seen that the distribution is anisotropic, i.e. specific directions increase the count-rate. This can be caused by strong radio active sources in the vicinity of the experiment apparatus. Therefore, the rate of random coincidences is dependent on the angle. The shown fit is of the form

$$N(\theta) = a + b \cdot \cos\left(2 \cdot \pi \frac{\theta}{360} - c\right) \tag{3}$$

wtih paramters $a = 0.153(14), b = -0.025(15)$ and $c = 110.87(70)$ and has a quality of $\chi^2/\text{ddof} = 1.57$, proving the function to be valid for modeling the distribution. Here it should be notice that 2π periodicity is enforced, since for a full rotation of the moving detector, the same signal is expected.

2.1.2 *Resolving Time and Single Detector Count Rates*

Another method is to calculate the accidental coincident rate with the resolving time and the count rate of the two detectors. For this the formula

$$N_{\text{acc}} = \sigma \cdot N_1 \cdot N_2 [1] \tag{4}$$

is used. Here we have the count rates N and the resolving time $\sigma = 2\tau$. The resolving time is set at the fast coincidence module to $\sigma = 30(4)$ ns. The large uncertainty does not come from used modules, but represents the range of the resolving time such that the coincidence is reasonable.

The count rates are shown in Figures 15 to 16

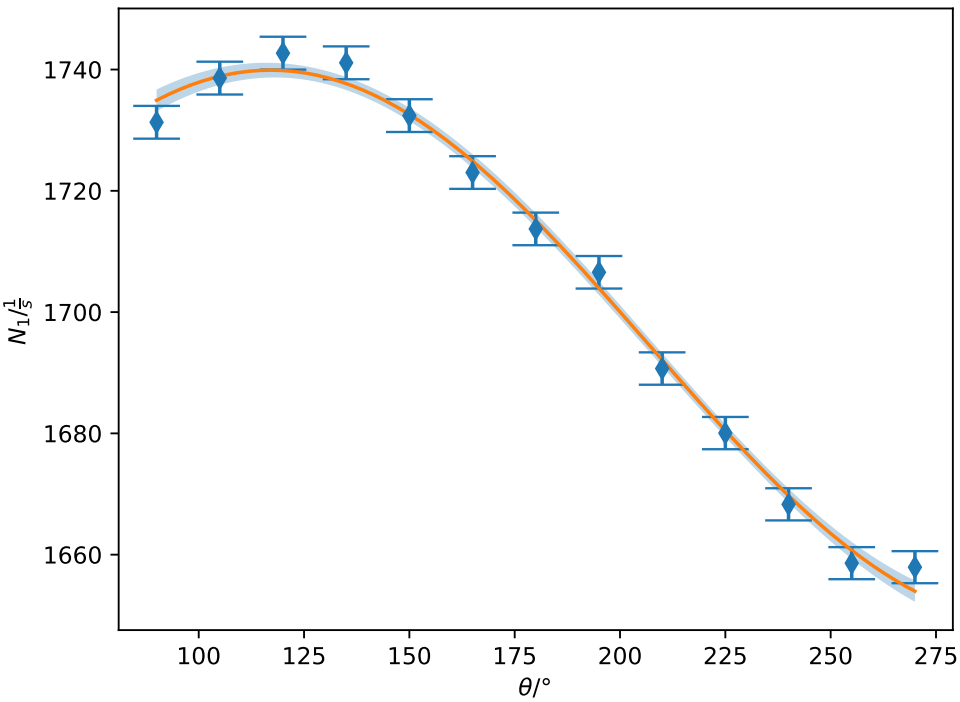

Figure 15: Count Rate for Detector 1 (Moving)

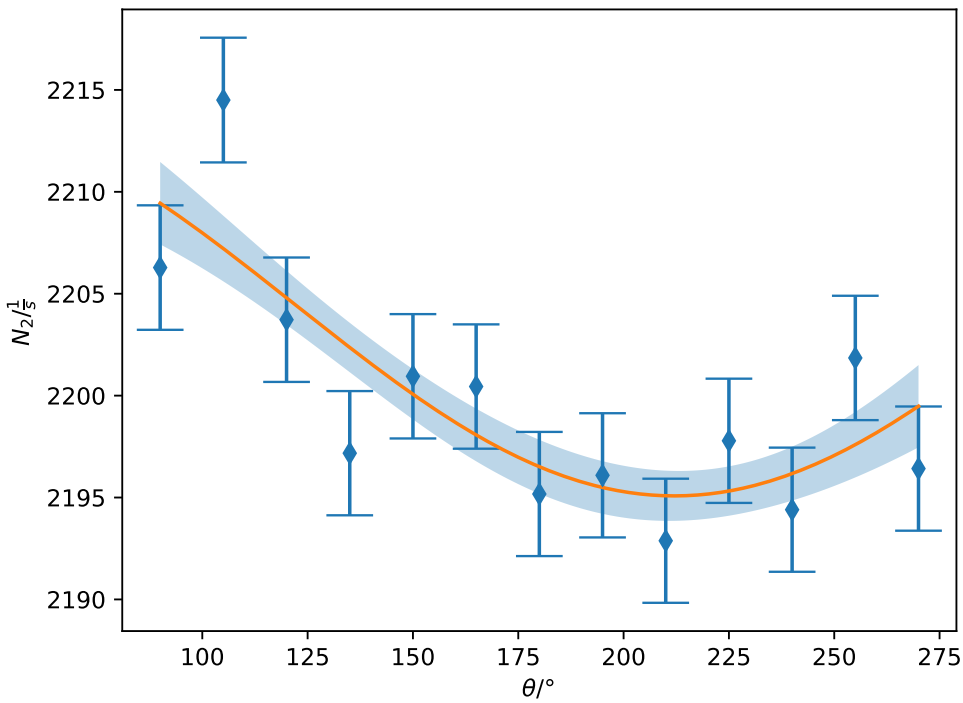

Figure 16: Count Rate for Detector 2 (Stationary)

As seen before in the direct measurement, an anisotropic distribution is observed in the moving detector. The quality $\chi^2/\text{ddof} = 0.97$ is well within a reasonable range with parameters $a = 1694.4(15), b = 45.5(13)$ and $c = 2.043(43)$. Optimally the quality should be 1. On the other hand, the quality $\chi^2/\text{ddof} = 1.52$ of the stationary detector is fairly high but also within a reasonable range wiht paramters $a = 2204.5(17), b = 9.4(22)$ and $c = 6.84(17)$. Even though the stationary detector should not be dependent on the angle, a dependency is still assumed, since the moving detector can screen of certain parts of the field of view of the stationary detector and thus influence the count rate. Here is also a 2π periodicity assumed. Using the Equation (4) again is probably not perfectly modeling the behaviour. A more appropriate Ansatz is an expansion in the spherical harmonics, where the data is fitted for the coefficients, since the spherical harmonics constitute a complete set. Taking another look at Equation (4) one might notice, that this is in fact already a expansion to first order in l and $m = 0$ of the spherical harmonics. This and the fact that the quality is close to 1, justifies still using Equation (4).

As a result of the count rates and resolution Figure 17 is obtained.

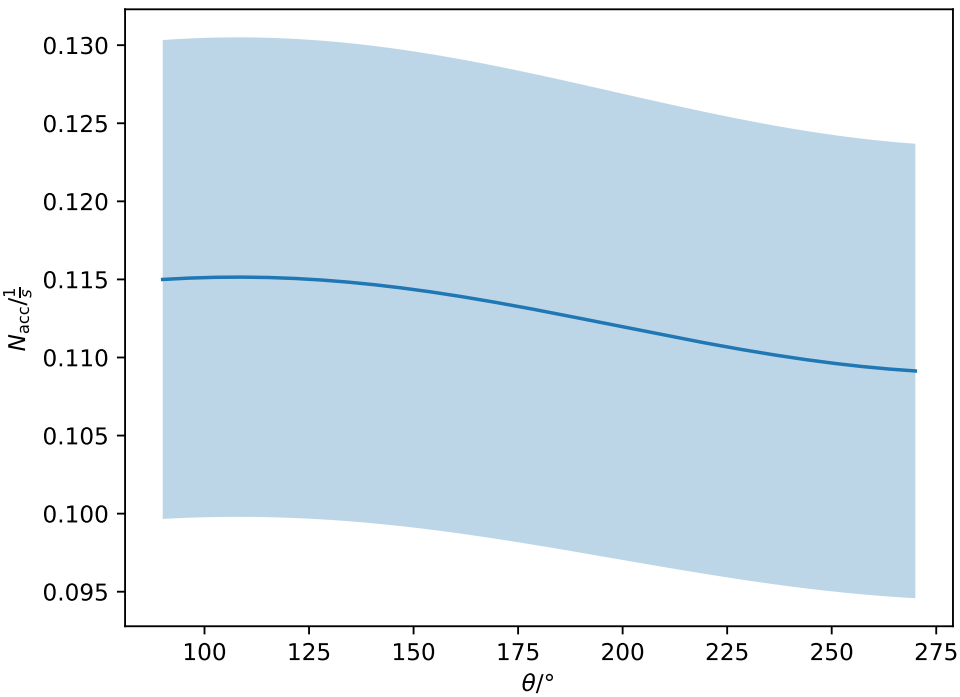

Figure 17: Accidental Coincidences Calculated through Resolving Time and Count Rates of Detector 1 and 2

For this the functions for $N_1(\theta)$ and $N_2(\theta)$ are used. As seen a small dependency on the angle is present. More significant though, is the large uncertainty, here seen from the large confidence band colored blue. So no real information beyond Gaussian distribution can be assumed. Thus the weighted mean using the original data points for the count rates is computed to an average of $\bar{N} = 0.1125(42)\,\frac{1}{s}$.

The calculated coincidence rate is with $4.8\,\sigma$ not very close to the directly measured accidental coincidence rate, which is ranging from 0.133 to 0.175 (see Figure 14). This deviation will stem from systematic errors, where the uncertainty on some measurements are underestimated or electronic noise could have triggered more coincidences in the fast branch, then would normally be triggered by external radiation that were originally planed to be measured by the random measurement.

2.2 *Stability Check*

To investigate the stability of the detectors over time, the main measurement is used to look at the fluctuations of the signals of the moving and stationary detector for constant angles over the multiple measurements for the different runs. Here fluctuations are defined as the standard deviation of of the signal over the different runs, since the runs happen at different points in time.

If the detectors would be stable over time, all fluctuations of the count rate would be within the uncertainty given by the Poisson distribution, i.e. the fluctuations could be caused by the uncertainty. Therefor at each angle the standard deviation and the uncertainty is calculated. Here the uncertainty is calculated as for every weighted mean like

$$\bar{\sigma}_x = \sqrt{\frac{1}{\sum_i \sigma_i^{-2}}} \tag{5}$$

To compare the different angles to each other, the standard deviation and the uncertainty are then divided by the mean of the count rates for each angle. This of course makes them only approximative comparable, but that should suffice for this analysis. As a result Figure 18 and Figure 19 are obtained for the moving detector and stationary detector respectively.

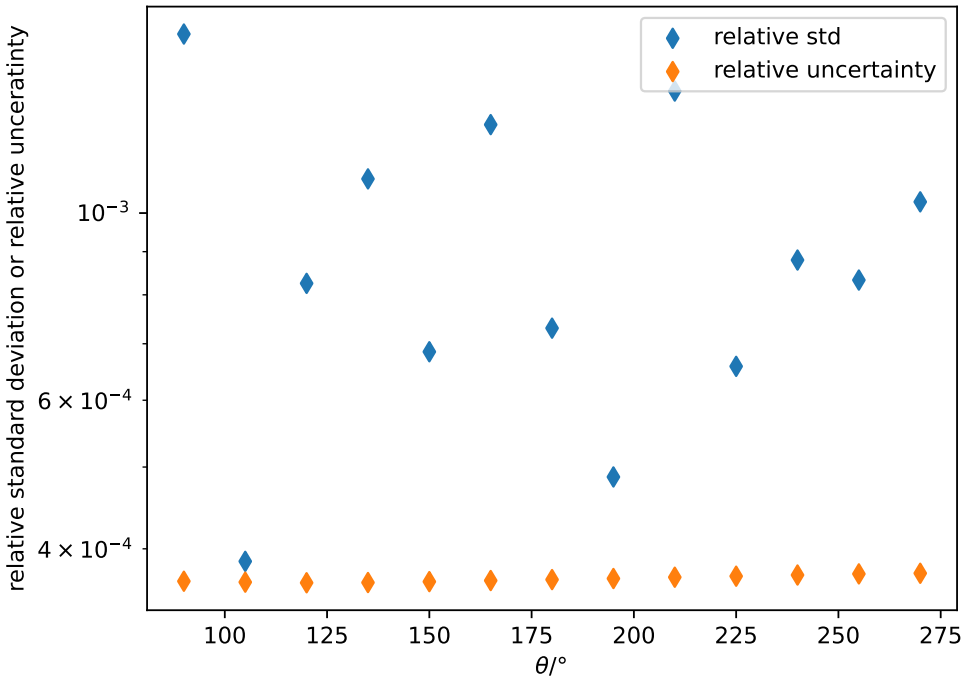

Figure 18: Relative Standard Deviation and Uncertainties of Count Rates for Constant Angles of the Moving Detector

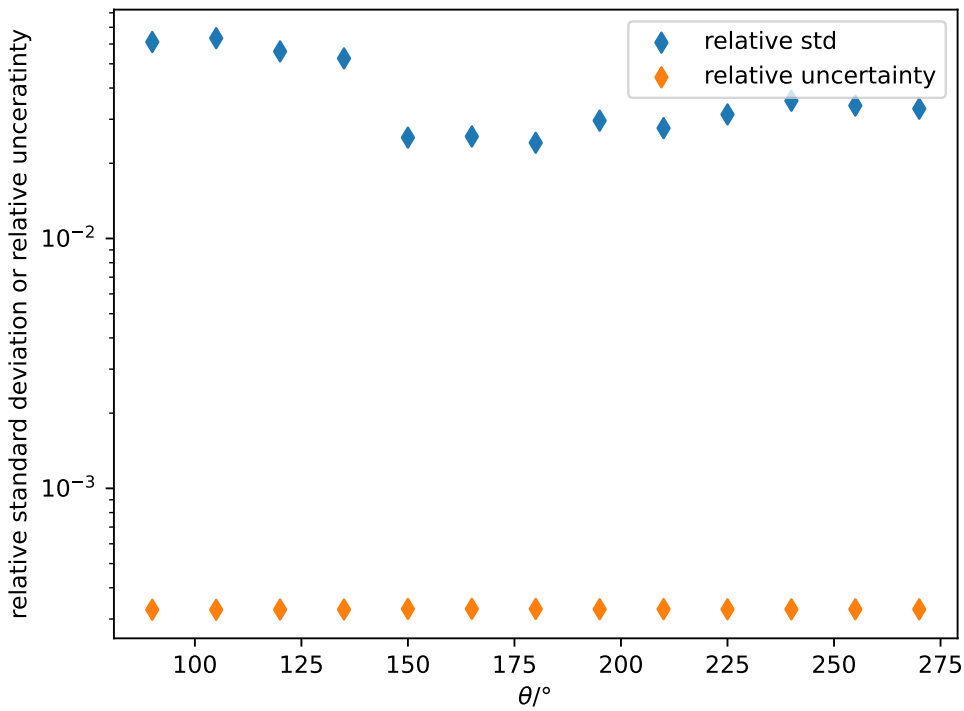

Figure 19: Relative Standard Deviation and Uncertainties of Count Rates for Constant Angles of the Stationary Detector

For the moving detector and all angles, the fluctuation is higher then the uncertainty. The detector is not stable over time and the counts depend on the time measured. The same is the case for the stationary detector. Therefore the count rate is dependent on the specific time. To get more insight into the behaviour of the time dependency, the count rates are shown in Figures 20 to 21. Here, over all runs are averaged. As always the weighted average is take, where the weight is the square of the inverse uncertainty.

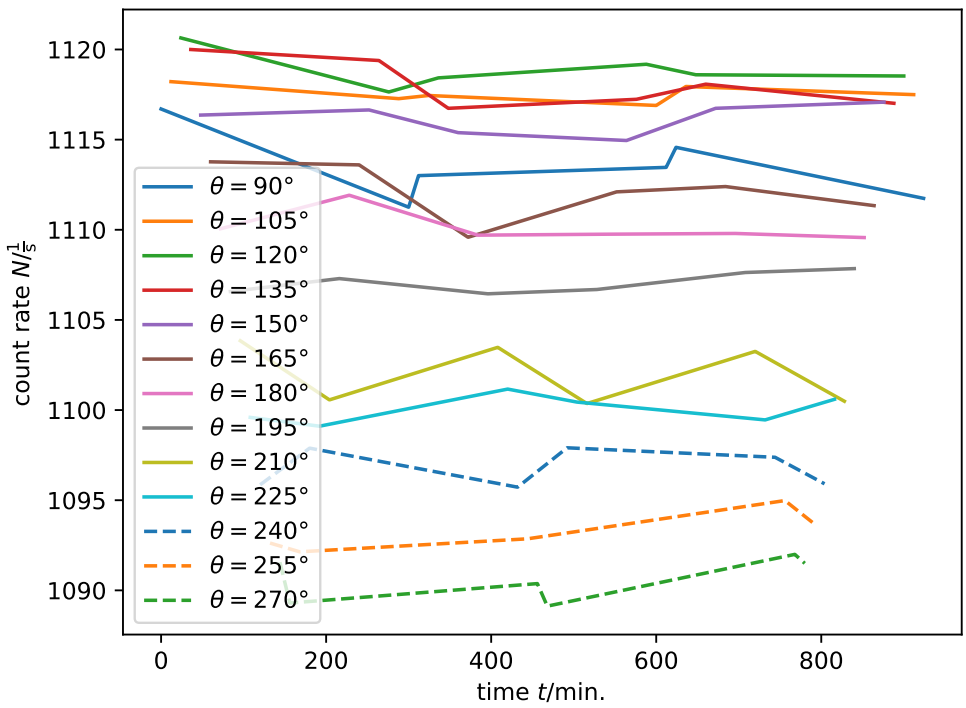

Figure 20: Count Rates for Constant Angles of Moving Detector over Time

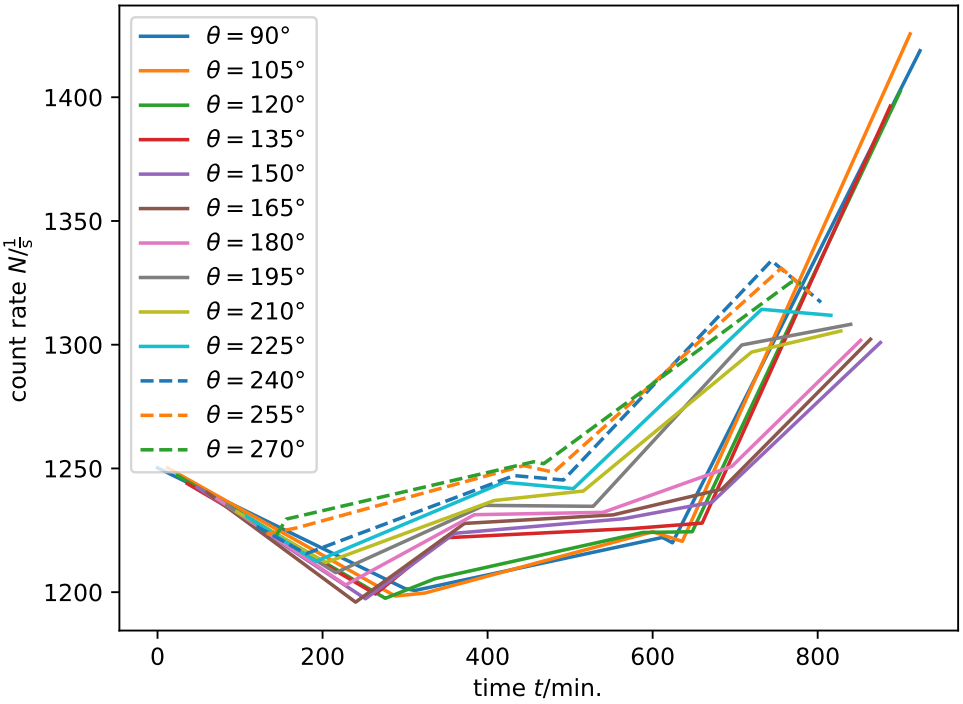

Figure 21: Count Rates for Constant Angles of Stationary Detector over Time

Against the usual manner, the data points are connected and uncertainties omitted. The purpose of this, is to enhance visibility for all 13 different angles, since only their general behaviour is of interest and not their exact values. From the figures themself it gets apparent that the count rate increases for all angles for the stationary detector, while this behaviour is even stronger for the first four angles. The moving detector has no such strong dependency on time.

The same procedure is now done with the total coincidence rate and with the fast branch coincidence rate instead of separating into detector 1 and 2, i.e. moving and stationary.

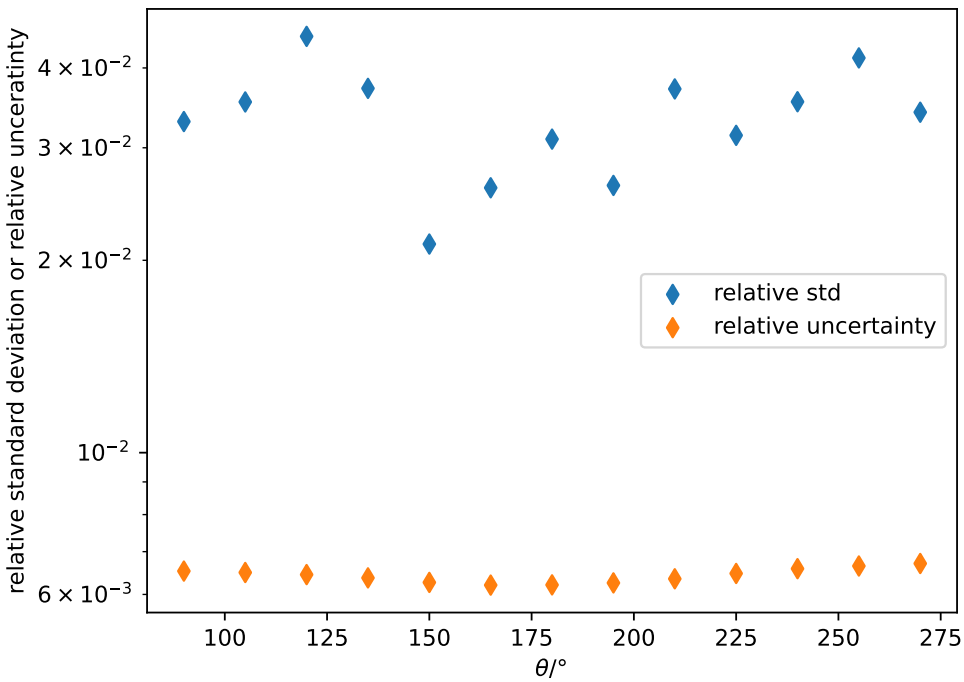

Figure 22: Relative Standard Deviation of Count Rates for Constant Angles for the Coincidence Rate

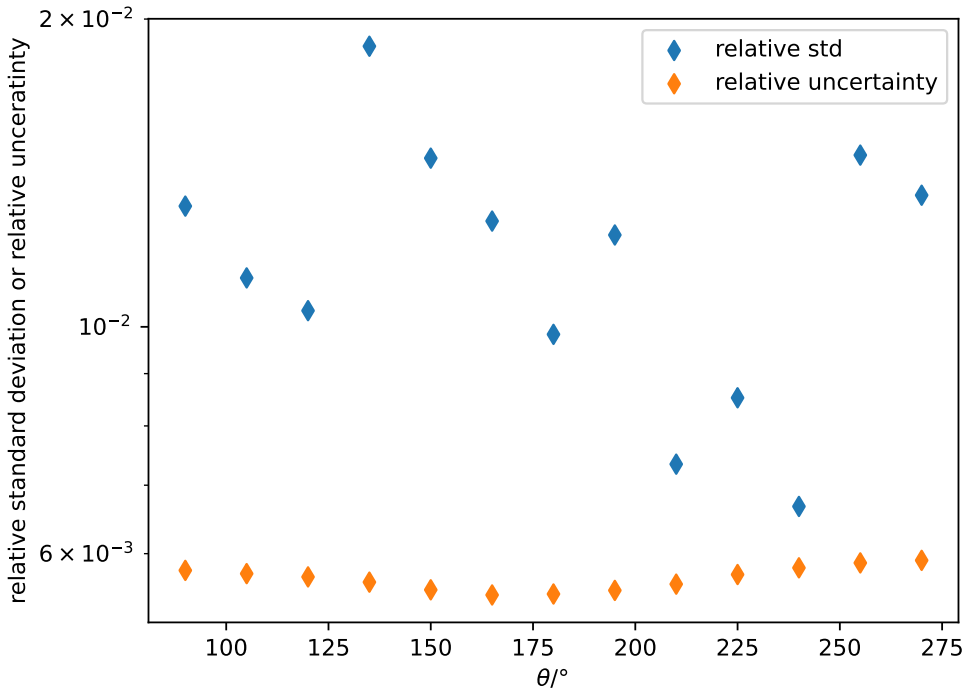

Figure 23: Relative Standard Deviation of Count Rates for Constant Angles of the Fast Branch Coincidence Rate

All fluctuations are above the uncertainty (see Figures 22 to 23), making them significant, i.e. non-neglectable. The count rates per angle over the measuring time are shown in Figures 24 to 25.

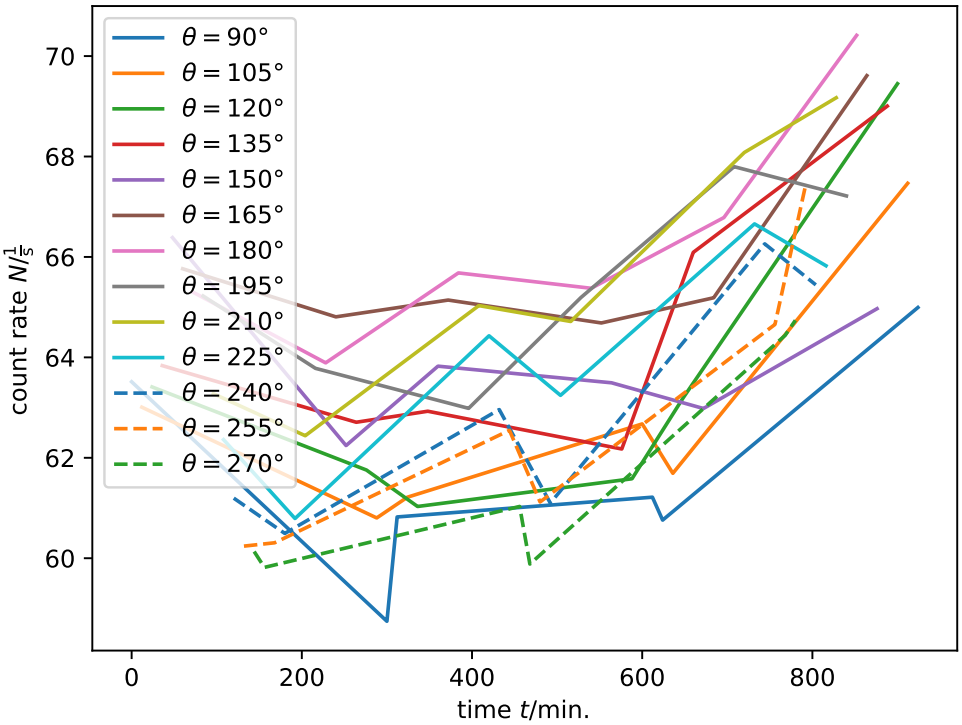

Figure 24: Coincidence Rate for Constant Angles over Time

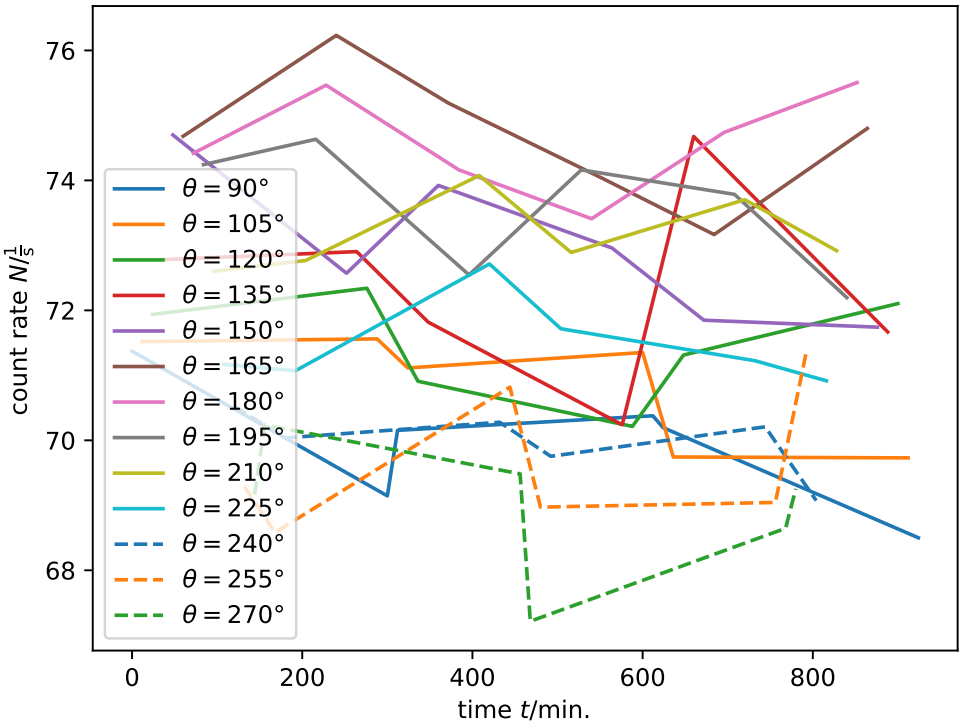

Figure 25: Fast Branch Coincidence Rate for Constant Angles over Time

The coincidence rate has a minimum around 300 minutes into the measurement for all angles, but increases again until the end to even higher coincidence rates then at the beginning of the measurement. The fast branch has no particular behaviour for the coincidence rates.

Overall, the time stability is poorly since the count rate is always dependent on the time when measured. Albeit this, the final result should not depend on this time dependency too strongly, since multiple measurements of the same angle are taken at different times, thereby should even out that time dependency.

To improve the measurement, it is advisable to make more runs and measure for less time per angle per run. For this measurement each angle was measured 12 minutes per run for a total of 6 runs and thus 936 minutes or 15 hours and 36 minutes. Doing e.g. 3 minutes per run for total of 24 runs would reduce the time dependency significantly, though taking the same amount. Of course the moving detector contributes a few extra seconds for repositioning but this should be neglectable. During the experiment the detector was noticed to take 2 to 4 seconds for moving.

2.3 *Correction for Random Coincidences*

From here on the random coincidences from the direct measurement in Section 2.1.1 are subtracted from the main measurement. The distinguishment between different angles is kept in case angular dependent photon sources skew the measurement.

2.4 *Misalignment Correction*

Considering that the source is not perfectly centered, its misalignment is investigated. For this an angle dependent correction factor

$$\alpha(\theta) = \frac{N(\theta)}{N(180^\circ)} \tag{6}$$

with

$$N(\theta; r_d, r_s, \theta_s) = \frac{N_0}{\Delta R(r_d, r_s, \theta, \theta_s)^2}, \tag{7}$$

$$\Delta R(r_d, r_s, \theta, \theta_s) = (\vec{R}(r_d, \theta) - \vec{R}(r_s, \theta_s))^2 \tag{8}$$

$$\vec{R}(r, \theta) = r \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix} \tag{9}$$

results in

$$\alpha(\theta) = \frac{r_d^2 - 2r_d r_s \cos(180^\circ - \theta_s) + r_s^2}{r_d^2 - 2r_d r_s \cos(\theta - \theta_s) + r_s^2} \tag{10}$$

The main measurement data is used to show this ratio α in Figure 26

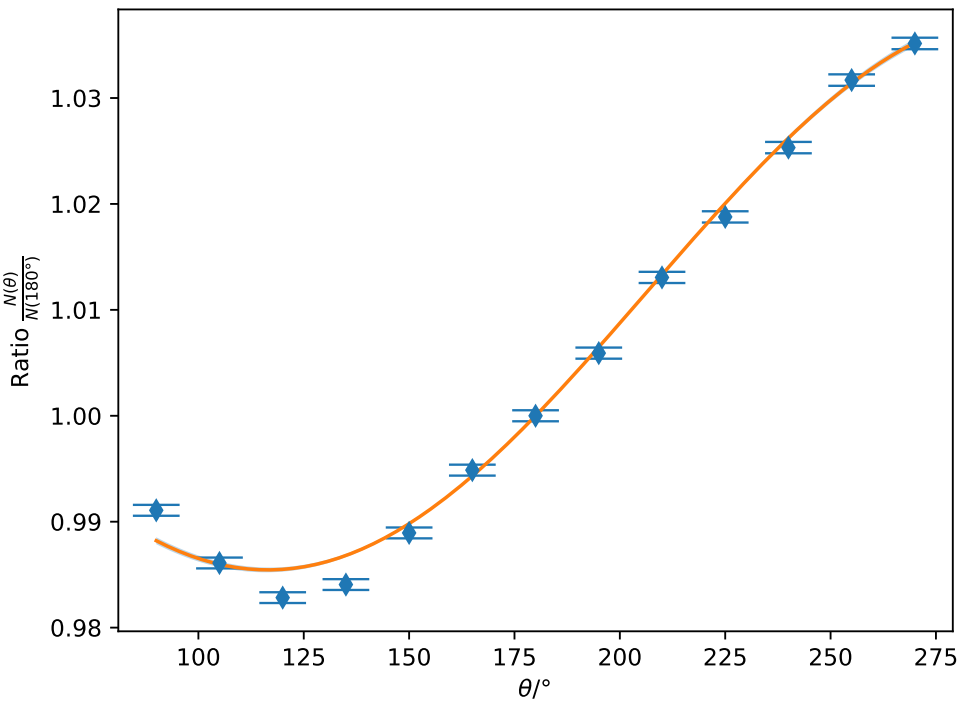

Figure 26: Misalignment Measurement

As a result the displacement $r_s = 64.9(13)\,\mu$m at angle $\theta_s = 116.5(14)°$ with a fit quality of $\chi^2/\text{ddof} = 9.82$ is found. The χ^2 is very high and shows that during the experiment and analysis larger systematic uncertainties have not been accounted for or wrong assumptions have been made.

This correction ratio $\alpha(\theta)$ can now be applied for the final analysis by multiplying the ratio with the count rates. It will correct of up to 3.5% as seen in Figure 26.

2.5 *Solid Angle Corrections*

Since the detector is not point-like but has a spacial volume, the incoming rays are not perfectly normal to the detector surface. Therefore a correction has to be made to coefficients

$$A_{ii} = \frac{A_{ii}^{\text{exp}}}{Q_{ii}(r_1)Q_{ii}(r_2)} \tag{11}$$

of the expansion of the angular correlation function

$$W(\theta) = N_0 \cdot (1 + A_{22}P_2(\cos\theta) + A_{44}P_4(\cos\theta)) \tag{12}$$

Since the in the previous section determined displacement of 64.9(13) μm is within the uncertainty of 1 mm of the authors ability to measure the distances, the source can be seen as centered. Additionally assuming that the detectors have the same distance to the source, Equation (11) can be simplified with $r_1 = r_2 = r = \frac{d}{2}$ where $d = 90(1)$ mm is the distance from detector 1 to detector 2. Additionally it i assumed that the scintillator is 5 mm deep into the detector, making the total distance 10(1) mm. Therefore the detectors are separated by 5 cm from the source. Numerical values are given for the correction factors where they depend on this distance and the energy of the photon [2]. Even though the observed transitions happen at 1173.2 keV and 1332.5 keV, the correction factors plateau beyond 1 MeV. Thus the correction factors $Q_{22}(r) = 0.9057$ and $Q_{44}(r) = 0.7113$ for photons of 1 MeV are used.

2.6 *Angular Correlation Function*

The count rates are now corrected by misalignment and random coincidences by multiplying with the correction factor $\alpha(\theta)$ and subtracting the random coincidences

$$N_{\text{corr}}(\theta) = N(\theta) \cdot \alpha(\theta) - N_{\text{rand}}(\theta) \tag{13}$$

Here the random coincidence function $N_{\text{rand}}(\theta)$ corresponds to Equation (3) with the parameters from the direct measurement of the random coincidences. As a result the distribution in Figure 27 is obtained.

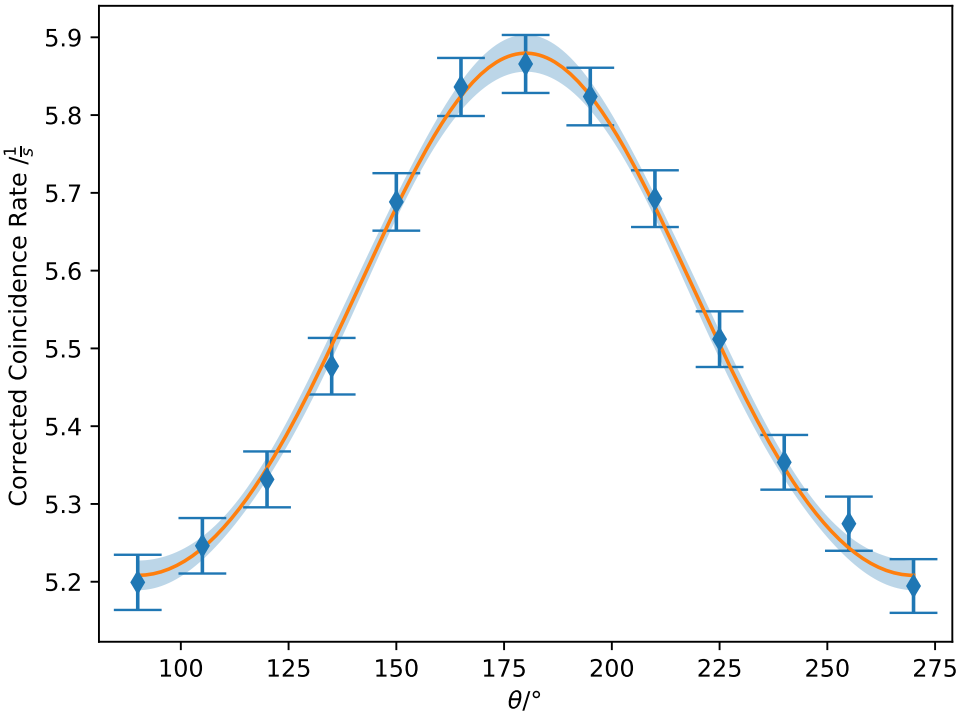

Figure 27: Main Measurement

Here the fit to the angular correlation function in Equation (12) is done. The quality $\chi^2/\text{ddof} = 0.22$ with parameters $N_0 = 5.411(11)$, $A_{22}^{\text{exp}} = 0.0799(37)$ and $A_{44}^{\text{exp}} = 0.0067(47)$ is achieved. The χ^2 is very small showing that at some point the uncertainties overestimated or plainly more data is needed to reduce the error. With the correction from the previous section the coefficients are adjusted to $A_{22} = 0.097(13)$ and $A_{44} = 0.0045(93)$. Given their large uncertainties, they lie within 1σ of the literature values $A_{22}^{\text{lit}} = 0.1020$ and $A_{44}^{\text{lit}} = 0.0091$ [2]. Especially $A_{44} = 0.0045(93)$ has an uncertainty larger then the value itself.

Last but not leas, other cascades are fitted to the experimental distribution. The chosen cascades and the fit results are in Figure 28 with Table 1. Here the coefficients $A_{ii} = F_i(L_1L_1I_iI) \cdot F_i(L_2L_2I_fI)$ for a $I_i(L_1)I(L_2)I_f$ cascade are set to literature values of the corresponding cascade[2] and only the normalisation N_0 is optimized.

cascade	χ^2/ddof	N_0
$0-1-0$	657.31	4.9661(92)
$0-2-0$	1647.61	4.4778(84)
$2-1-0$	12.22	5.4537(99)
$4-1-0$	255.81	5.553(10)
$4-2-0$	0.28	5.4087(98)

Table 1: Quality for Normalisation Constant N_0 Fitted to Different Cascades

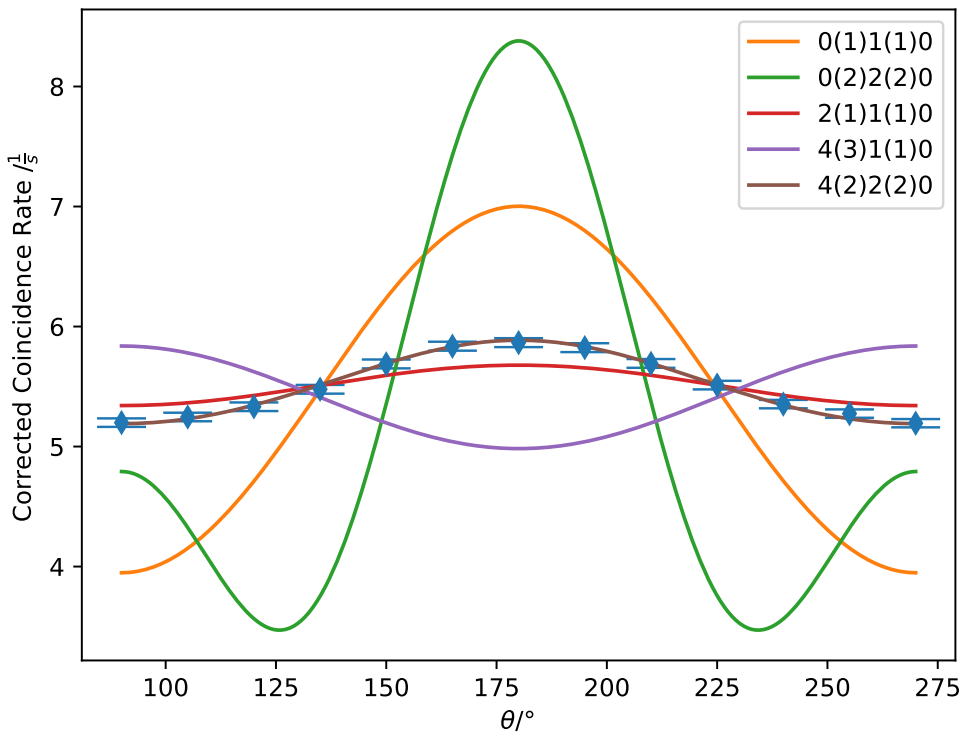

Figure 28: Different Cascades Fitted to the Experimental Angular Distribution

As seen, the assumed $4-2-0$ is most likely correct since the quality is here best, even though the χ^2 is quite low, which shows, that more data is needed.

3 Conclusion

After collecting the data from the random and main measurement, the main analysis started. For this the random coincidence rate was determined by directly measuring the coincidence, ranging from ($1.3\,\frac{1}{s}$ to $1.75\,\frac{1}{s}$) and by calculating it with the single count rates of the moving and stationary detector and the resolving time ($\bar{N} = 0.1125(42)\,\frac{1}{s}$.

Following the stability is measured by comparing the over-time-fluctuations with the uncertainty. There it is seen that the measurement is not stable over time. The influence on the date should still remain marginal though, each angle was measured at multiple time points.

Correcting for misalignment as been done by multiplying by a factor corresponding to an increment in count rate by being closer to the source for specific angles. Here it could also be determined that the source is off-center by 64.9(13) μm at an angle of 116.5(14)° from the stationary detector.

The last correction made is caused by the spacial volume of the used detector. Therefor the expansion coefficients in the angular correlation function have been corrected by a factor corresponding to a distance of 50 mm and already known from literature[2].

Finally, the angular correlation is plotted and fitted to the expansion. There coefficients $A_{22}^{\text{exp}} = 0.0799(37)$ and $A_{44}^{\text{exp}} = 0.0067(47)$ are measured. They are within 1σ of the literature values $A_{22}^{\text{lit}} = 0.1020$ and $A_{44}^{\text{lit}} = 0.0091$ due to their large uncertainty. Also, other cascades have been considered. There it could bee seen that the presumed 4-2-0 cascade is most likely.

References

[1] Leo, W.R.: Techniques for Nuclear and Particle Physics Experiments: A How-to Approach, 2nd edn. Springer, Berlin (1994)

[2] Siegbahn, K.: Alpha-, Beta-, and Gamma-Ray Spectroscopy vol. 2, p. 1695. North-Holland Publishing Company, Amsterdam (1965)